JAMIU DAMILARE DAHUNSI

Cyber-Physical Energy Systems Security & Resilience
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RESEARCH FOCUS

Cybersecurity and resilience of cyber-physical energy systems; secure OT/IT connectivity for sensing and control (PMUs, metering, DER-facing networks); intrusion-aware monitoring and adaptive defenses for smart grids and industrial control; embedded/edge security; red-team / blue-team reasoning for critical infrastructure.

EDUCATION

M.S., Electrical and Computer Engineering, Carnegie Mellon University (GPA: 3.63/4.00)

Energy / cyber-physical systems Core Courses: Applied Smart Grid Telecommunications (A), Photovoltaic Systems Engineering (A) Electrical Power Principles, Basic Electrical Engineering 1 and 2.

Cybersecurity, systems, and AI (defense of CPS/ICS environments): Applied Cryptography (A); Network Security (A−); Ethical Hacking (A); Information Security; Deep Learning (A); Data Inference and Applied Machine Learning; Agentic AI; Cloud Computing; Wireless Networks.

Graduate Teaching Assistant: Wireless Networks and Applications (Prof. Peter Steenkiste), Spring 2026.

B.Eng., Telecommunications Engineering, Federal University of Technology Minna, Nigeria 2018–2023

CGPA: 4.20/5.00 (Second Class Honours, Upper Division; WES unofficial ≈ 3.57/4.00). Energy & systems coursework: Electric Power Principles; Electrical Machines; Control Engineering; Measurements & Instrumentation; plus full telecommunications/networking curriculum.

RESEARCH EXPERIENCE — ENERGY CPS / OT SECURITY

Research Assistant,Carnegie Mellon University / Kigali Collaborative Research Center Summer 2026–Present

Cyber-secure Connectivity for C37.2 and OpenPMU Ecosystem Phasor Measurement Units | UKRI-funded | Supervisor: Prof. Barry Rawn

- Design and implement cyber-secure end-to-end connectivity for OpenPMU server/RTU assets to lab hosts, campus VMs (wired/wireless), and client systems on Rwanda Energy Group OT networks.
- Co-develop connectivity requirements spanning laboratory systems, industrial partner equipment, and EV-fleet related infrastructure; collaborate with KCRC, Rwanda Energy Group, Ampersand, and DieDev.

Research Assistant, ECE, Carnegie Mellon University (Prof. Theophilus Benson) 2025–Apr 2026

- Connectivity and outage-impact modelling for networked infrastructure; dependency analysis relevant to resilience of critical networked systems.

Technical Instructor CMU Africa Bridge Program (Rwanda) 6 weeks

- Trained undergraduates on Arduino Nano 33 BLE + Edge Impulse TinyML: sensing, model training, and on-device deployment for constrained embedded platforms (edge intelligence for CPS).

PUBLICATION (CYBER-PHYSICAL ENERGY IOT)

J. D. Dahunsi et al., "Design and Construction of Cloud-Based Energy Metering, Monitoring and Control System," IEEE NIGERCON 2024; IEEE Xplore, Mar. 2025. DOI: 10.1109/NIGERCON62786.2024.10926972

End-to-end cyber-physical energy system: metering, remote monitoring, and control over networked/cloud infrastructure—integrity and availability of energy data and actuation paths.

PROFESSIONAL EXPERIENCE — NETWORK & CRITICAL-INFRASTRUCTURE SECURITY

Network Engineer, Zone Payment Network Apr 2024–Present

- Architect and operate security-hardened production networks for financial institutions (critical infrastructure class): IPsec/GRE tunnels, Palo Alto / Check Point / StrongSwan, firewalls, IDS/IPS, 802.1X, RADIUS/TACACS; hybrid cloud (AWS/Azure) and on-prem.

- Maintain high availability (target >99.5%) with SolarWinds monitoring, automated backups, and incident-ready operations under real adversarial and reliability pressure.
- Secure introduction of agentic AI into production infrastructure: least-privilege access, secrets hygiene, monitoring, and blast-radius limits for autonomous actions—practical blue-team discipline for AI-augmented operations.

Network Engineer, Huawei Technologies Nigeria Jan 2024–Jun 2025

- Designed and deployed carrier/enterprise SDN and IP networks (MPLS, IPv6, QoS, wireless controllers) for multi-site clients including energy-sector environments reliability and security under operational constraints.
- First Prize, Huawei ICT Competition Global Finals 2023–2024 (Network Technology) Naational then global championship, Shenzhen.

Systems Administrator, Summitech Computing Nov 2022–Dec 2023

- Secure systems operations, hardening, and infrastructure reliability for production clients.

Network Security Intern, ipNX Nigeria Limited Jan 2022–Nov 2022

- Network security operations in ISP/enterprise environments: monitoring, access control, incident support.

TEACHING & MENTORSHIP

- Graduate TA, Wireless Networks and Applications, Carnegie Mellon University (Spring 2026).
- Cybersecurity trainer / TA: FUT Minna and NITDA IT Hub (2020–2023)—hands-on security for undergraduates and early-career professionals.
- CMU Africa Bridge: embedded TinyML instructor for undergraduate cohort in Rwanda.

COMPETITIVE DISTINCTIONS

- First Prize — Huawei ICT Competition Global Finals 2023–2024, Network Technology (Shenzhen).
- First Prize — Huawei ICT Competition Nigeria National Finals 2023, Network Technology.
- First Prize — SIGCOMM POD Hackathon 2025 (Carnegie Mellon University Africa).

CERTIFICATIONS (INDUSTRY-VALIDATED SECURITY & NETWORKING)

CCNP Enterprise, HCIP Datacom; HCIA Security; HCIA WLAN; HCIA Datacom, Microsoft AZ-900; Microsoft SC-900, Fortinet Certified Associate in Cybersecurity

TECHNICAL SKILLS

Cybersecurity / OT-IT: network & system security, Applied Cryptography, Ethical Hacking / offensive assessment mindset, VPN/IPsec, firewalls, IDS/IPS, zero-trust access controls, secure hybrid-cloud networking.
Energy CPS adjacency: smart-grid telecommunications, PV systems engineering, OpenPMU / C37.2 secure connectivity, energy metering & control IoT.
AI/ML for security & edge: Python, PyTorch, TinyML / Edge Impulse, anomaly-aware and agentic-AI deployment security.
Systems: Linux, Docker, SDN/IP, Wireshark, SolarWinds; learning HIL / real-time simulation stacks as needed for SENTRY/ANGLE.